



Tutorial Letter 103/0/2026

Introduction to Programming I COS1511

Year Module

Department of Computer Science

IMPORTANT INFORMATION

Please register on myUnisa, activate your myLife e-mail account and make sure that you have regular access to the myUnisa module website, COS1511-2026-Y, as well as your group website.

Note: This is a fully online module. It is, therefore, only available on myUnisa.

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1 INTRODUCTION

Dear Student

The following instructions relate to assignment number 02 and 03.

We wish you every success with your assignment!

2 ASSIGNMENT

2.1 Outcomes covered in the assignment

The following outcomes will be covered in:

Assignment number 02

Conditional execution

- If statements
- While
- Boolean values
- Nested if statements
- Switch statements
- While loops
- For loops
- Nested loops

Assignment number 03:

Functions

- Writing functions
- Local and global variables
- Void functions
- Reference parameters

Data structures

- One-dimensional arrays
- Arrays as parameters
- Two-dimensional arrays

2.2 Assignment information

Details of assignment can be found in the table below:

Assignment weight contribution towards final mark:	See Tutorial letter 101
Method of assessment (project/takehome/written/portfolio)	written
Total marks:	
Assignment 2	25
Assignment 3	45
The assignment is: (compulsory/mandatory/elective)	Yes
The assignment will/not be proctored using IRIS	no
Turn-it-in will be used for projects/takehome/written assignments	no

The date for submission can be found on myUNISA.

2.3 Assignment submission instructions

- Unisa, as a comprehensive open distance e-learning institution (CODeL), is moving towards becoming an online institution. You will therefore see that all your study material, assessments and engagements with your lecturer and fellow students will take place online. We use myUnisa as our virtual campus.
- The myUnisa virtual campus will offer students access to the myModules site, where learning material will be available online and where assessments should be completed. This is an online system that is used to administer, document, and deliver educational material to students and support engagement between academics and students.
- The myUnisa platform can be accessed via <https://my.unisa.ac.za>. Click on the myModules 2024 button to access the online sites for the modules that you are registered for.
- The university undertakes to communicate clearly and as frequently as is necessary to ensure that you obtain the greatest benefit from the use of the myModules learning management system. Please access the announcements on your myModules site regularly, as this is where your lecturer will post important information to be shared with you.
- When you access your myModules site for the module/s you are registered for, you will see a welcome message posted by your lecturer. Below the welcome message you will see the assessment shells for the assessments that you need to complete. Some assessments may be multiple choice, some tests, others written assessments, some forum discussions, and so on. All assessments must be completed on the assessment shells available on the respective module platforms.
- To complete quiz assessments, please log on to the module site where you need to complete the assessment. Click on the relevant assessment shell (Assessment 1,

Assessment 2, etc.). There will be a date on which the assessment will open for you. When the assessment is open, access the quiz online and complete it within the time available to you. Quiz assessment questions are not included in this tutorial letter (Tutorial Letter 101) and are only made available online. You must therefore access the quiz online and complete it online where the quiz has been created.

- It is not advisable to use a cell phone to complete the quiz. Please use a desktop computer, tablet or laptop when completing the quiz. Students who use a cell phone find it difficult to navigate the **Online Assessment** tool on the small screen and often struggle to navigate between questions and successfully complete the quizzes. In addition, cell phones are more vulnerable to dropped internet connections than other devices. **If at all possible, please do not use a cell phone for this assessment type.**
- For written assessments, please note the due date by which the assessment must be submitted. Ensure that you follow the guidelines given by your lecturer to complete the assessment. Click on the submission button on the relevant assessment shell on myModules. You will then be able to upload your written assessment on the myModules site of the modules that you are registered for. Before you finalise the upload, double check that you have selected the correct file for upload. Remember, no marks can be allocated for incorrectly submitted assessments.

2.4 Assignment questions

2.4.1 Assignment 2

Question 1

[5]

In South Africa, the pension fund contribution for employees is 7.5% of your pensionable salary towards the fund, while your employer's pension contribution is 13% of your pensionable salary.

For example, an employee earns R3, 500 is pensionable salary. Therefore, an employee contribution will be as follows ($7.5\% \times R3, 500$), which is R262.50, While an employer pays ($13\% \times R3, 500$), which is R455.

Write a C++ program to calculate the total pension fund contribution for the employee and the employer.

Question 2

[20]

Income tax rates are divided into seven segments commonly known as income tax brackets. All taxpayers pay increasing income tax rates as their income rises through these segments. If you're trying to determine your marginal tax rate or your highest federal tax bracket, you'll need to know two things:

- **Income tax rate:** The various percentages at which taxes are applied on individual income tax brackets: The ranges of income to which a tax rate applies (currently there are four as shown above).

Yearly income	Percentage
<= 250000	10%
> 250000 and <= 500000	15%
>500000 and <= 1000000	20%
> 1000000	25%

2.1. Write a C++ program that prompts the user to enter the taxable income per year and computes the income tax for a year.

- **Your filing status:** The filing status options are to file as single, married filing jointly, married filing separately, or qualified widow.

2.2. Write a C++ switch statement to determine each case of the filing status.

Case	Status	Calculation
1	Single	Use individual tax rates applied in 2.1
2	Married filing jointly	(income = income1 + income2)
3	Married filing separately	Use individual tax rates applied in 2.1
4	Qualified widow	(Income = Income – 100000)

2.4.2 Assignment 3

[45]

Question 1

[23]

BMI or **Body Mass Index** is a method of estimating a person's **body fat** levels based on a person's weight and height measurement. The BMI is calculated by dividing the person's body mass/weight (in kilograms) by the square of the body height (in metres). The result is universally expressed in units of kg/m².

BMI formula: BMI = weight /(height)²

BMI	Weight status
Below 18.5	Underweight
18.5-24.9	Healthy
25.0-29.9	Overweight
30.0 and above	Obese

Write a program that calculates a person's **BMI** and displays the weight status according to the BMI value, indicating whether the person is underweight, healthy, overweight or obese. The BMI weight statuses and related values are listed in the above table.

The program should have the following functions:

- A) **getData** that prompts and returns their weight and height; (5)
- B) **calcBMI** to calculate the BMI ; (5)
- C) **displayFitnessResults** to display the BMI and relevant weight status message; (8)
- D) **main** function (5)

Question 2

[10]

The *A+Driving School* wants a C++ program to determine whether a student passes an examination. Anyone who wants to learn to drive at the school has to write an examination paper consisting of 25 multiple choice questions. The student's answers are stored in an array called **answerArray**. The correct answers are stored in another array, namely **correctArray**. A student passes if 13 or more of his answers agree with the correct answers. You have to write a **bool** function, called **passOrNot**, to determine whether the student passes or not. The value **true** should be returned if the student passes and the value **false** should be returned if the student does not pass.

Assume the following:

- a declaration of a global constant:

```
const int NUM_QUESTIONS = 25; // number of questions
```
- three declarations in the main function:

```
char answerArray[NUM_QUESTIONS]; // student's answers
char correctArray[NUM_QUESTIONS]; // correct answers
bool pass;
```
- values have been assigned already to all the elements of the two arrays
- the function is called in the main function as follows:

```
pass = passOrNot(answerArray, correctArray);
```

Write down the complete function **passOrNot**.

Question 3

[12]

Suppose we have two very similar two-dimensional arrays of type **int**, namely **matrixA** and **matrixB**, both with 5 rows and 6 columns. For every row of the arrays we have to determine how many of the corresponding elements are not the same, i.e. we have to count how many

times the integer in a specific position in the specific row of **matrixA** is not identical to the integer in the corresponding position (in the corresponding row) of **matrixB**.

- A) Declare two constants, namely **NUM_ROWS** and **NUM_COLUMNS**, where the number of rows (i.e. 5) and the number of columns (i.e. 6) of the two-dimensional arrays are stored. (2)
 - B) Declare two two-dimensional arrays, **matrixA** and **matrixB**, in terms of **NUM_ROWS** and **NUM_COLUMNS**. (2)
 - C) Assume that values have been assigned to all the elements of both arrays already. For each row, determine how many times the value in the same positions of **matrixA** and **matrixB** are not identical and display the number. Assume that the following declaration has been made:

```
int nrNotSame ; // nr of non-identical values
```

 (8)
-

2.5 Proctoring and plagiarism

Since 2020 Unisa conducts all its assessments online. Given stringent requirements from professional bodies and increased solicitations of Unisa's students by third parties to unlawfully assist them with the completion of assignments and examinations, the University is obliged to assure its assessment integrity through the utilisation of various proctoring tools: Turnitin, Moodle Proctoring, the Invigilator App and IRIS. These tools will authenticate the student's identity and flag suspicious behaviour to assure credibility of students' responses during assessments. The description below is for your benefit as you may encounter any or all of these in your registered modules:

Turnitin is a plagiarism software that facilitates checks for originality in students' submissions against internal and external sources. Turnitin assists in identifying academic fraud and ghost writing. Students are expected to submit **typed** responses for utilisation of the Turnitin software.

The **Moodle Proctoring tool** is a facial recognition software that authenticates students' identity during their Quiz assessments. This tool requires access to a student's **mobile or laptop camera**. Students must ensure their camera is activated in their browser settings prior to their assessments.

The **Invigilator "mobile application-based service** does verification" of the identity of an assessment participant. The Invigilator Mobile Application detects student dishonesty-by-proxy and ensures that the assessment participant is the registered student. This invigilation tool requires students to download the app from their Play Store (Google, Huawei and Apple) on their **mobile devices** (camera enabled) prior to their assessment.

IRIS Invigilation software verifies the identity of a student during assessment and provides for both manual and automated facial verification. It has the ability to record and review a student's assessment session. It flags suspicious behaviour by the students for review by an academic

administrator. IRIS software requires installation on students' **laptop devices** that are enabled with a webcam.

Students who are identified and flagged for suspicious dishonest behaviour arising from the invigilation and proctoring reports are referred to the disciplinary office for formal proceeding.

Please note:

Students must refer to their module assessment information on their myModule sites to determine which proctoring or invigilation tool will be utilised for their formative and summative assessments.

3 ACADEMIC DISHONESTY

3.1 Plagiarism

Plagiarism is the act of taking the words, ideas and thoughts of others and presenting them as your own. It is a form of theft. Plagiarism includes the following forms of academic dishonesty:

- Copying and pasting from any source without acknowledging the source.
- Not including references or deliberately inserting incorrect bibliographic information.
- Paraphrasing without acknowledging the original source of the information.

3.2 Cheating

Cheating includes, but is not limited to, the following:

- Completing assessments on behalf of another student, copying the work of another student during an assessment, or allowing another student to copy your work.
- Using social media (e.g. WhatsApp, Telegram) or other platforms to disseminate assessment information.
- Submitting corrupt or irrelevant files, this forms part of examination guidelines
- Buying completed answers from so-called "tutors" or internet sites (contract cheating).

3.3 For more information about plagiarism, follow the link below:

<https://www.unisa.ac.za/sites/myunisa/default/Study-@-Unisa/Student-values-and-rules>

4 IN CLOSING

Good luck!!